

University of St. Gallen

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Stochastic Methods in Finance

Assignment – Spring Semester 2026

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1 Introduction

- Graph of appl (Andrea)
- Small introduction (Andrea)

2 Building the Binomial Tree

- graph of small tree only prices (Mathis)
- explain how we build the tree (Juliette)

We build the binomial tree as follows. The starting point is the closing price of Apple on April 28, 2026, denoted S_0 . At each period, the price either goes up (multiplied by u) or down (multiplied by d), where $\Delta t = T/n = 0.02$ years. Both factors are calibrated using the volatility σ estimated from Apple's daily log-returns over 2020–2026. The larger σ , the larger u and the smaller d .

Starting from S_0 , each node splits into two branches: an up-branch ($S \times u$) and a down-branch ($S \times d$). After 25 periods, the tree contains $2^{25} \approx 33$ million possible paths. At each node, we store the cumulative sum $C_t = \sum_{j=0}^t S_j$, needed to compute the arithmetic average $\bar{S}_n = C_n/(n+1)$.

- explain u and d (Mathis)
- explain how we compute the stock price at each node (Mathis)
- link number of nodes to number of periods $1+t$ (Juliette)

At each period t , a recombining tree has $t+1$ distinct price nodes. At $t = 25$, this gives 26 distinct prices. However, since the Asian option payoff depends on the arithmetic average $\bar{S}_n$, two paths reaching the same final price can yield different payoffs. We therefore track all $2^{25} \approx 33$ million paths separately, even though they converge to only 26 distinct prices at maturity.

3 Arithmetic Average at Terminal Nodes

- add formula for average (Mathis)
- add tree with average (Mathis)
- Explain how we got these results (Juliette)
- add distributions (2x2 graph): C (distribution + Paths), $\bar{S}$ (distribution), S (distribution) (Mathis)
- Graphs descriptions (Andrea)
- calculate skewness and kurtosis of these distributions (Erwan)

4 Payoff at Terminal Nodes

- explain the payoff (Andrea)
- add formula (Andrea)
- Explain how we got these results (Juliette)
- Tree picture (With the payoff at the terminal nodes) (Mathis)
- Stats (done by Juliette)
- Payoff distribution (Terminal Nodes) (Mathis)
- Description and interpretation of the payoff distribution (Andrea)

Table 1 reports descriptive statistics of the payoff $A_T = \max(S_n - \bar{S}_n, 0)$ computed at the terminal nodes of the binomial tree ($t = 25$).

Statistic	Value (\$)
Count	3.36×10^7
Mean	15.71
Std	24.71
Min	0.00
25%	0.00
Median	0.00
75%	24.65
Max	321.58

Table 1: Descriptive statistics of the payoff A_T at the terminal nodes ($t = 25$).

The distribution of A_T is highly right-skewed: the median is zero, meaning more than half of all terminal paths yield a worthless option, i.e. $S_n < \bar{S}_n$. Notably, the standard deviation (24.71) exceeds the mean (15.71), reflecting the high dispersion of payoffs. Most paths expire worthless while a small number of highly favourable paths drive the mean upward. The maximum payoff of 321.58 corresponds to the path where the stock price increases at every period.

5 Risk-Neutral Probabilities

- intro why we need q (Erwan)
- proof of q (Erwan)
- explain relation with vol and interest rate (Erwan)
- Matrix vol interest rate (Mathis)
- Interpretation of q matrix (Erwan)
- show result of q (Erwan)

- Distribution of the payoff under q vs p (Juliette)
- Description of the differences between the two distributions (Juliette)

6 Backward Induction and Option Price

- explain backward induction depends on q and how we get the result (Mathis)
- graph of the tree with values (Mathis)
- Presentation of our result (Juliette)
- show the price of the option being the same with different paths (Erwan)
- compare with MC (Andrea)

7 Robustness Check

- matrix with different q same as earlier before (Mathis)
- Compare price with MC 2std fixed strike (Andrea)
- Compare the different scenarios (Andrea)

8 Normal Approximation

- Values distribution (Mathis)
- Kurtosis and skewness of the distribution (Erwan)

9 Conclusion

- summary of the results (Andrea)
- limitations of the model (Andrea)
 - comparaison with MC (Andrea)
- possible extensions (Juliette)
 - more iterations, more periods... (Juliette)
- final remarks (Erwan)